

ADA-Marker Answer Sheet

Anonymized local demo

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PROBLEM

Q1

Answer

3. (1) For example, 1c0111 then Z value will be 600111, it means the max len from the head to the current position can match with the string begins from the current position. #
- (2) To calculate (S, n) , we use left and right as a window. If it matches with Z-Algorithm def then record the length, set the right to the end of the match to create a window. and there will be three cases:
- In the window: use the former recorded Z value to write $O(1)$
 - cut of the window: just calculate brutally $O(n)$
 - In the window but former record not enough: We take the former record and continue calculate 接下来的字串 $O(n)$
- (3) $\dots \overset{\text{right}}{O(n)} - \overset{\text{left}}{O(n)} + \overset{\text{right}}{O(1)} - \overset{\text{left}}{O(1)} = \overset{\text{calculate 接下来的字串}}{O(n)} \#$
- (4) The correctness of the Algo is in the three cases, when $i \leq \text{right}$ we use the record. When $i > \text{right}$ we directly calculate. when i is near right or at the end of the window we use linear time to take record and do more calculation, hence we've included all the cases. proved #
- (5) We put string like this $S \times T$, and we calculate the Z value: $O(n+m)$. we examine the place Z value $> \text{len}(S)$ to choose the max: $O(m) \cdot O(n)$. #